La Scuola d'Italia / Dr. Huson / IB Math 12
13 October 2026

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## 1.10 Pre-Quiz: Statistics

*Review for Wednesday's quiz. Use your GDC. Give answers exactly, or correct to three significant figures. Write down the values you put into the calculator. An outlier is a value more than* $1.5 \times IQR$ *below* $Q_1$ *or more than* $1.5 \times IQR$ *above* $Q_3$*.*

1. [Maximum mark: 8]

   The table shows the number of hours, $h$, that 60 students spent exercising last week.

| Hours, $h$ | $0 \leq h < 2$ | $2 \leq h < 4$ | $4 \leq h < 6$ | $6 \leq h < 8$ | $8 \leq h < 10$ |
|---|---|---|---|---|---|
| Number of students | 6 | 14 | 20 | 12 | 8 |

   (a) State whether the data are discrete or continuous. [1]

   (b) Write down the modal class. [1]

   (c) Find an estimate of the mean number of hours. [2]

   (d) Explain why your answer to part (c) is an estimate. [1]

   (e) Find an estimate of the standard deviation. [1]

   (f) The 60 students are a stratified sample, by grade, of the 480 students in the school. There are 112 students in Grade 12. Find the number of Grade 12 students in the sample. [2]

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2. [Maximum mark: 9]

Fifteen runners recorded their times, in minutes, for a 5 km race.

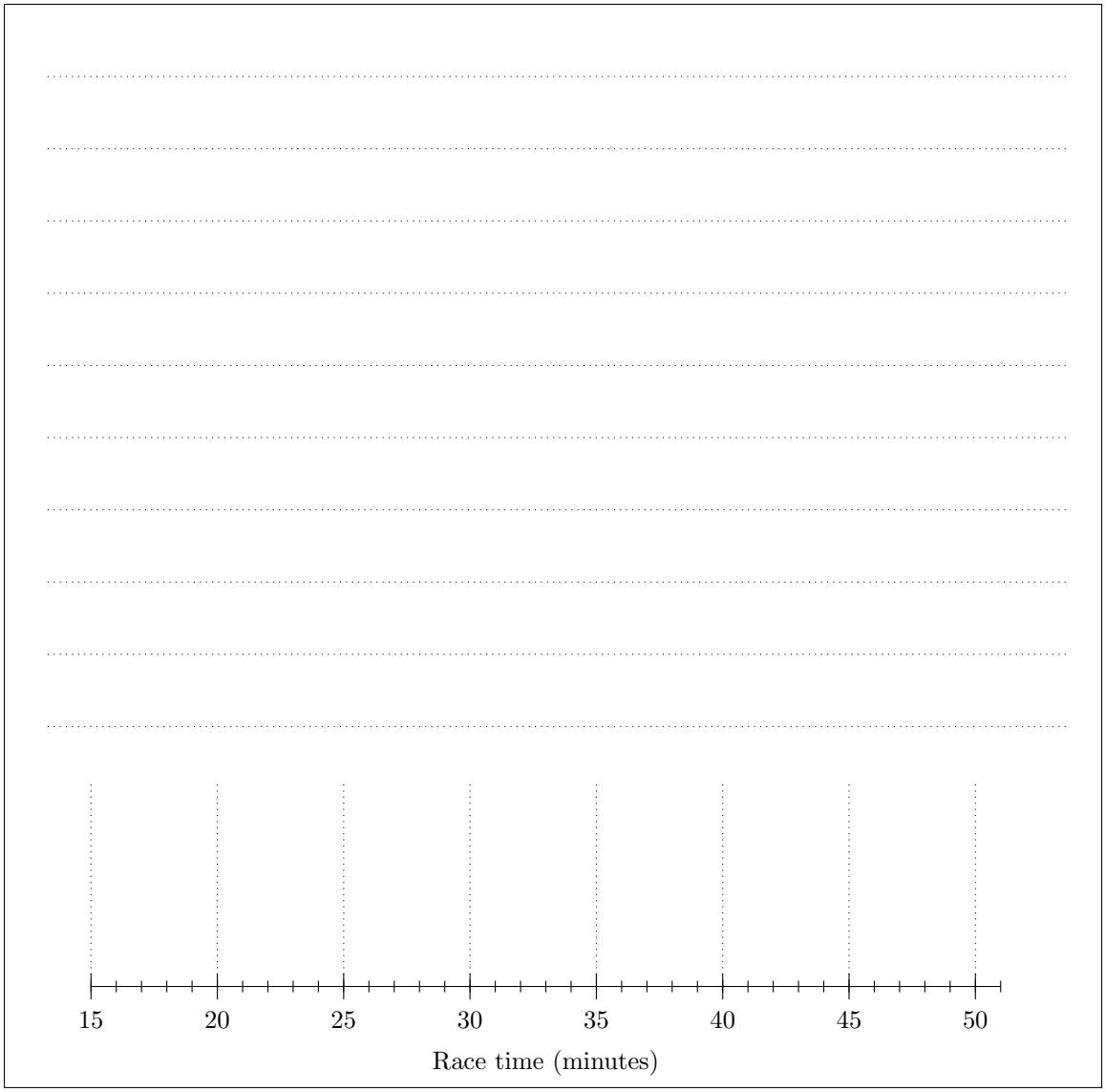
24, 27, 22, 31, 25, 29, 26, 23, 28, 30, 21, 26, 25, 44, 27

(a) Write down the median, the lower quartile $Q_1$ and the upper quartile $Q_3$. [2]

(b) Find the interquartile range. [1]

(c) Show that 44 is an outlier. [3]

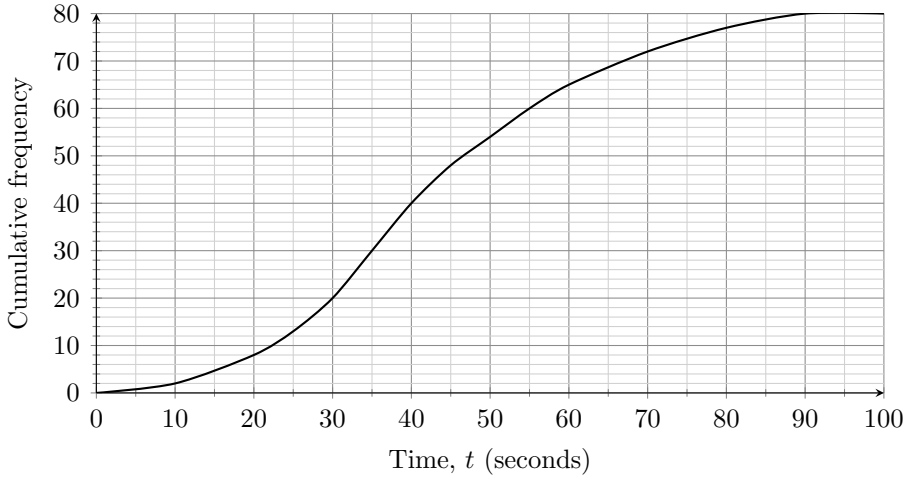
(d) On the scale at the bottom of the box, draw a box-and-whisker plot of the race times. Mark the outlier with a cross. [3]



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3. [Maximum mark: 9]

The cumulative frequency graph shows the time, $t$ seconds, that 80 students took to solve a puzzle.




(a) Write down the median time. [1]

(b) Find the interquartile range. [2]

(c) Find the number of students who took more than 60 seconds. [2]

(d) The 10% of students with the shortest times receive a prize. Find the greatest time a student could take and still receive a prize. [2]

(e) The times are grouped in the table below. Find the values of $p$ and $q$. [2]

| Time, $t$ (seconds) | $0 < t \leq 20$ | $20 < t \leq 40$ | $40 < t \leq 60$ | $60 < t \leq 100$ |
|---|---|---|---|---|
| Number of students | 8 | $p$ | $q$ | 15 |

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4. [Maximum mark: 9]

The table shows the number of hours, $x$, that eight students spent revising for a test, and their test scores, $y$ (out of 100).

| Hours revising, $x$ | 2 | 3 | 5 | 6 | 8 | 9 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|
| Test score, $y$ | 48 | 55 | 59 | 66 | 70 | 78 | 80 | 88 |

(a) Find Pearson's product-moment correlation coefficient, $r$. [1]

(b) Describe the correlation between $x$ and $y$. [2]

(c) Write down the regression line of $y$ on $x$, in the form $y = ax + b$. [2]

(d) Interpret the meaning of the value of $a$ in this context. [1]

(e) Use your regression line to estimate the test score of a student who revised for 10 hours. [2]

(f) Explain why the regression line should not be used to estimate the test score of a student who revised for 20 hours. [1]